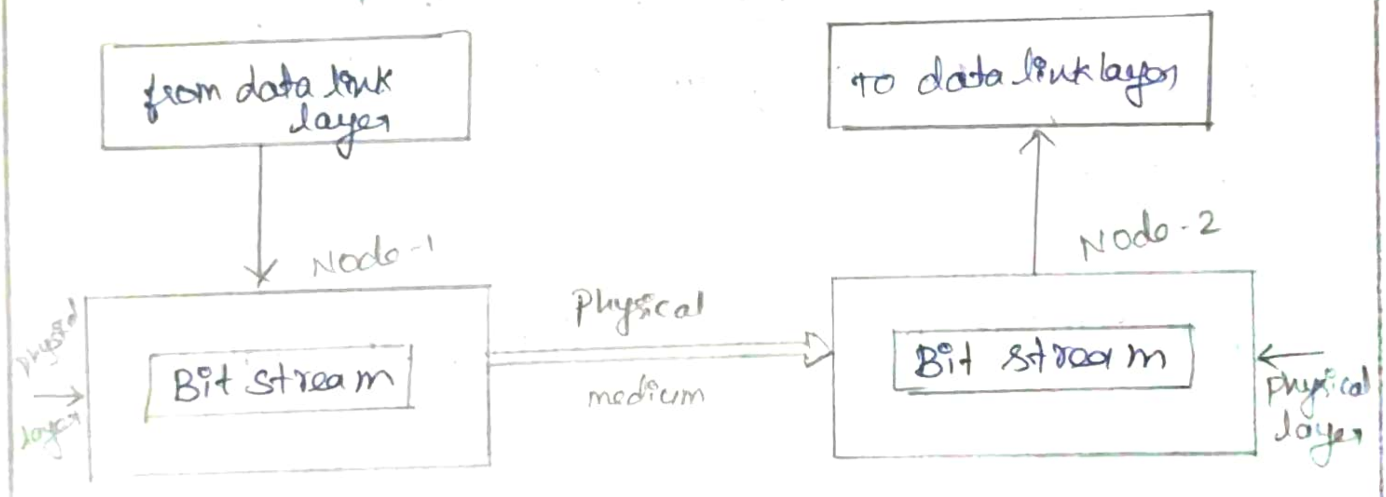
1) Explain in detail about the layer in OSI model.

Layers in OSI models:

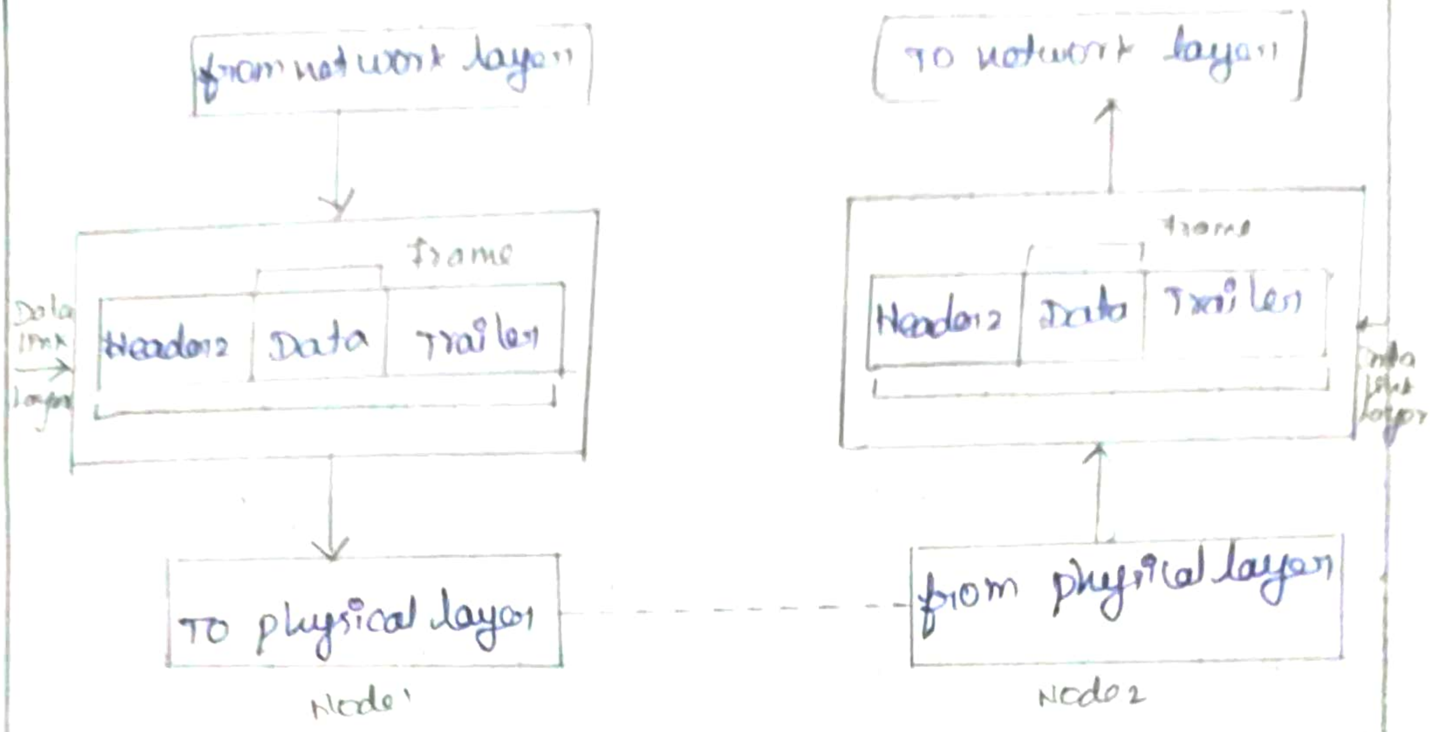
1. physical layer:

Physical layer is the lowest layer of the OSI model. physical layer coordinates the functions required to transmit a bit stream over a communication channel. It deals with electrical and mechanical specifications of interface and transmission media.



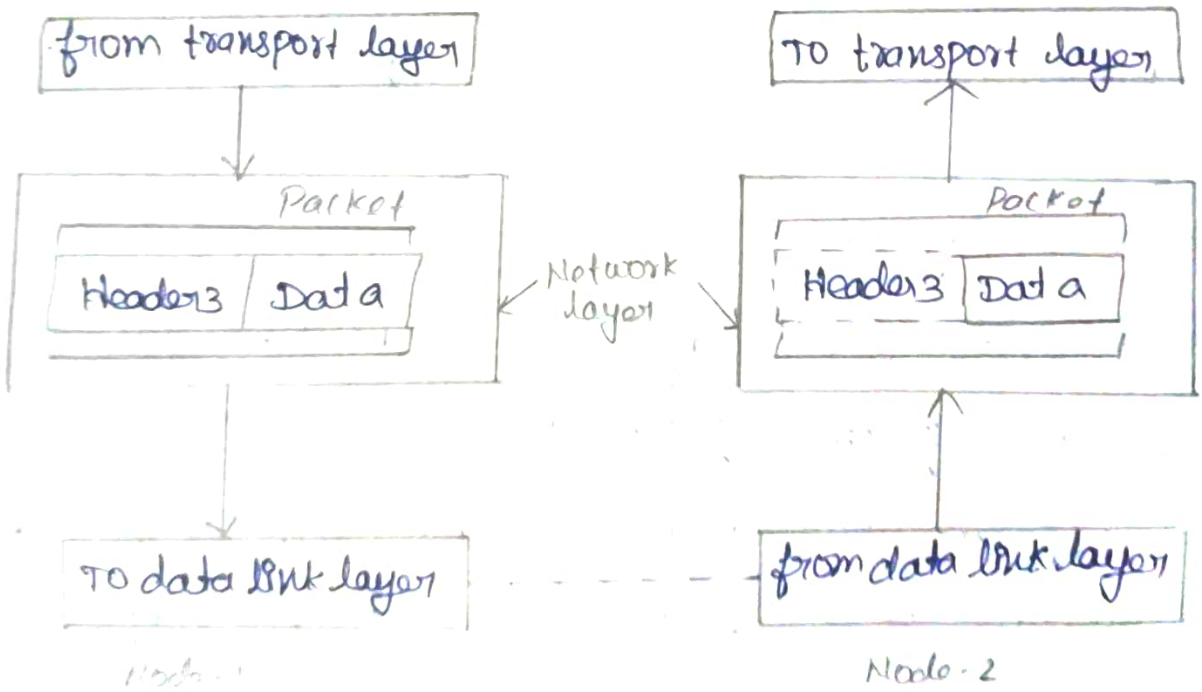
2. Data link layer:

The Data link layer is responsible for transmitting frames from one node to the next. It transform the physical layer to a reliable link making it an error free link to upper layer.



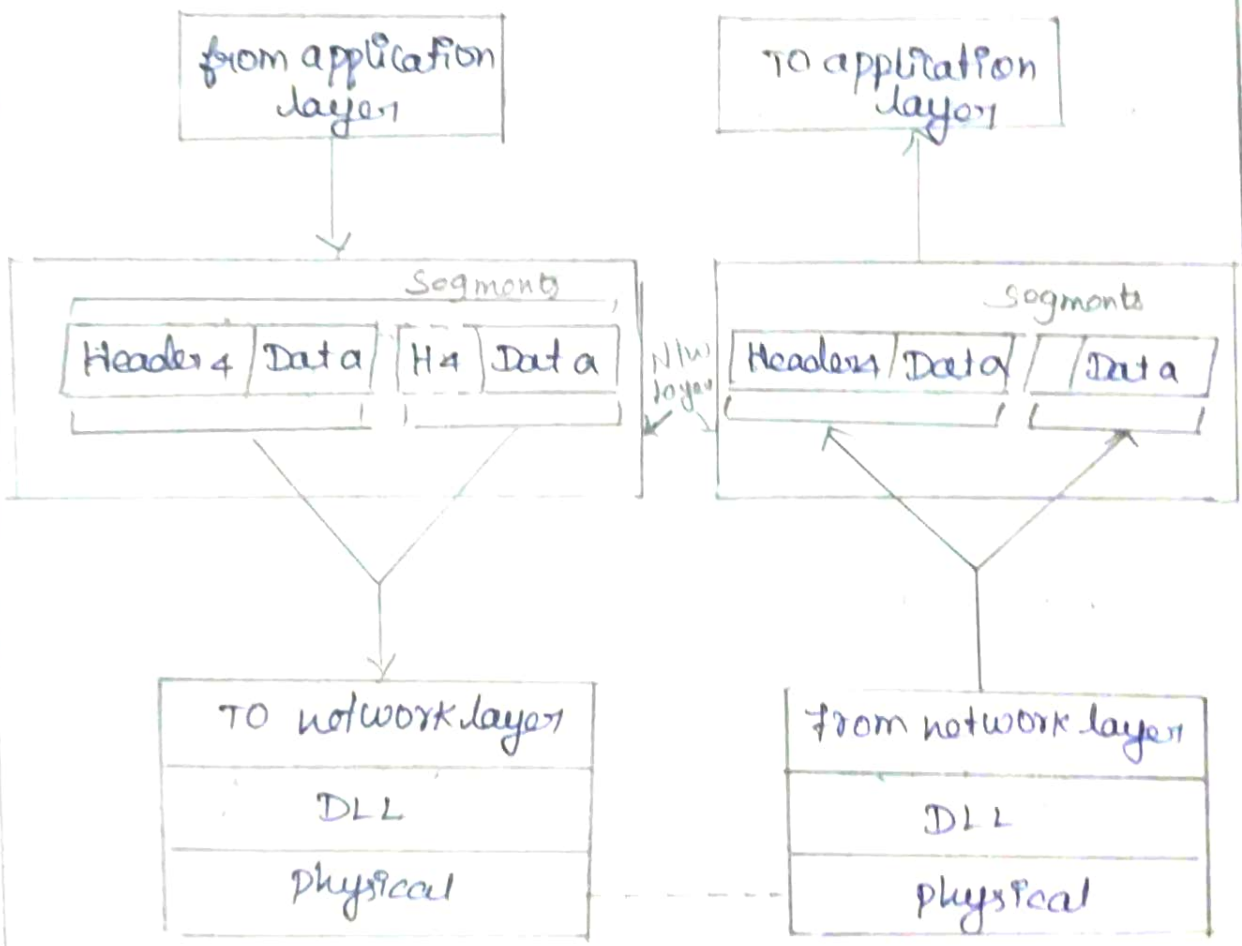
3. Network layer:

The network layer is responsible for the delivery of packets from the source to destination.



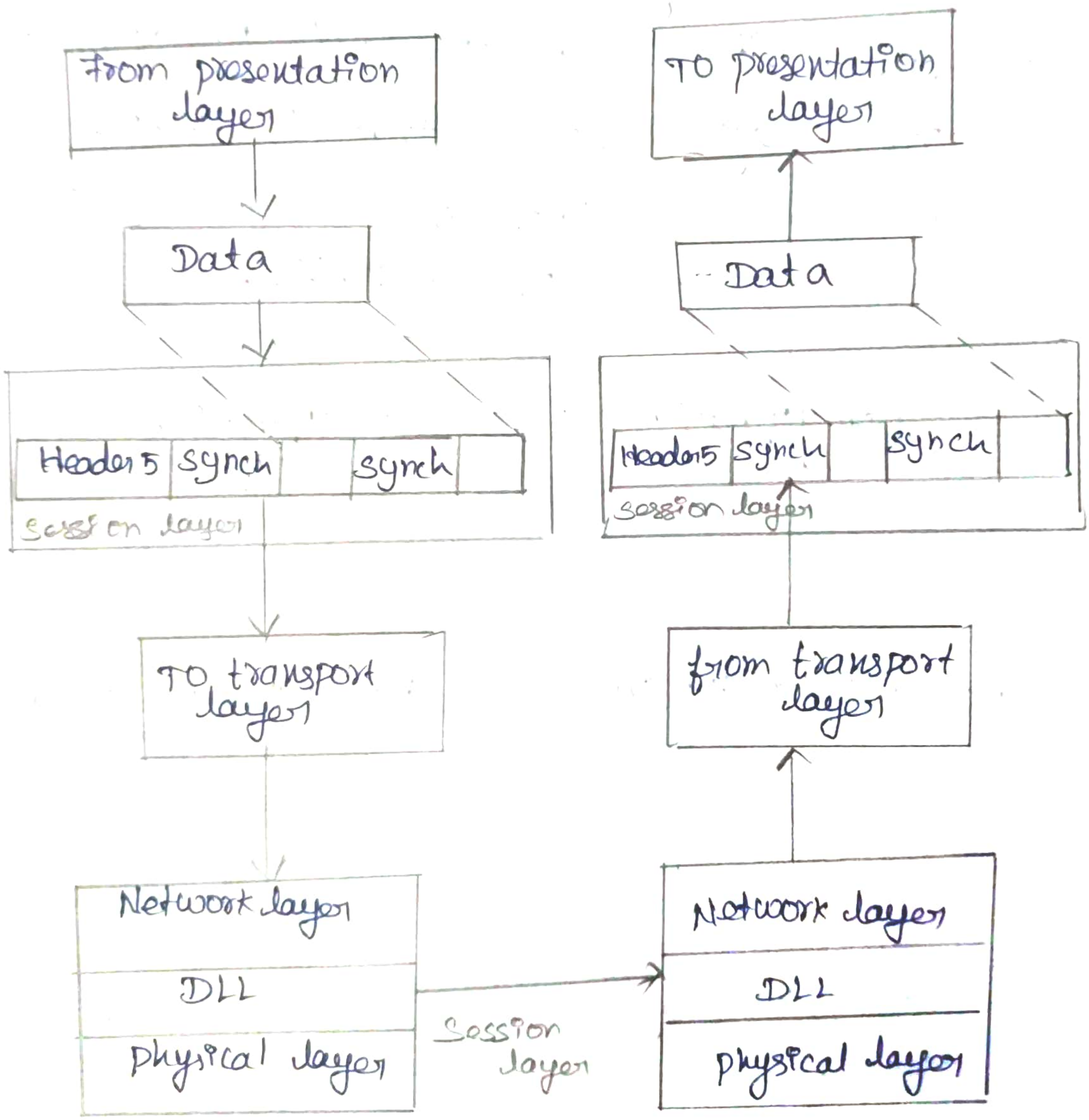
4. Transport layer:

The transport layer is responsible for delivery of message from one process to another. The network that host to destination delivery of individual packets considering it as independent packet. But transport layer ensures that the whole message arrives intact and in order with error control and process control.



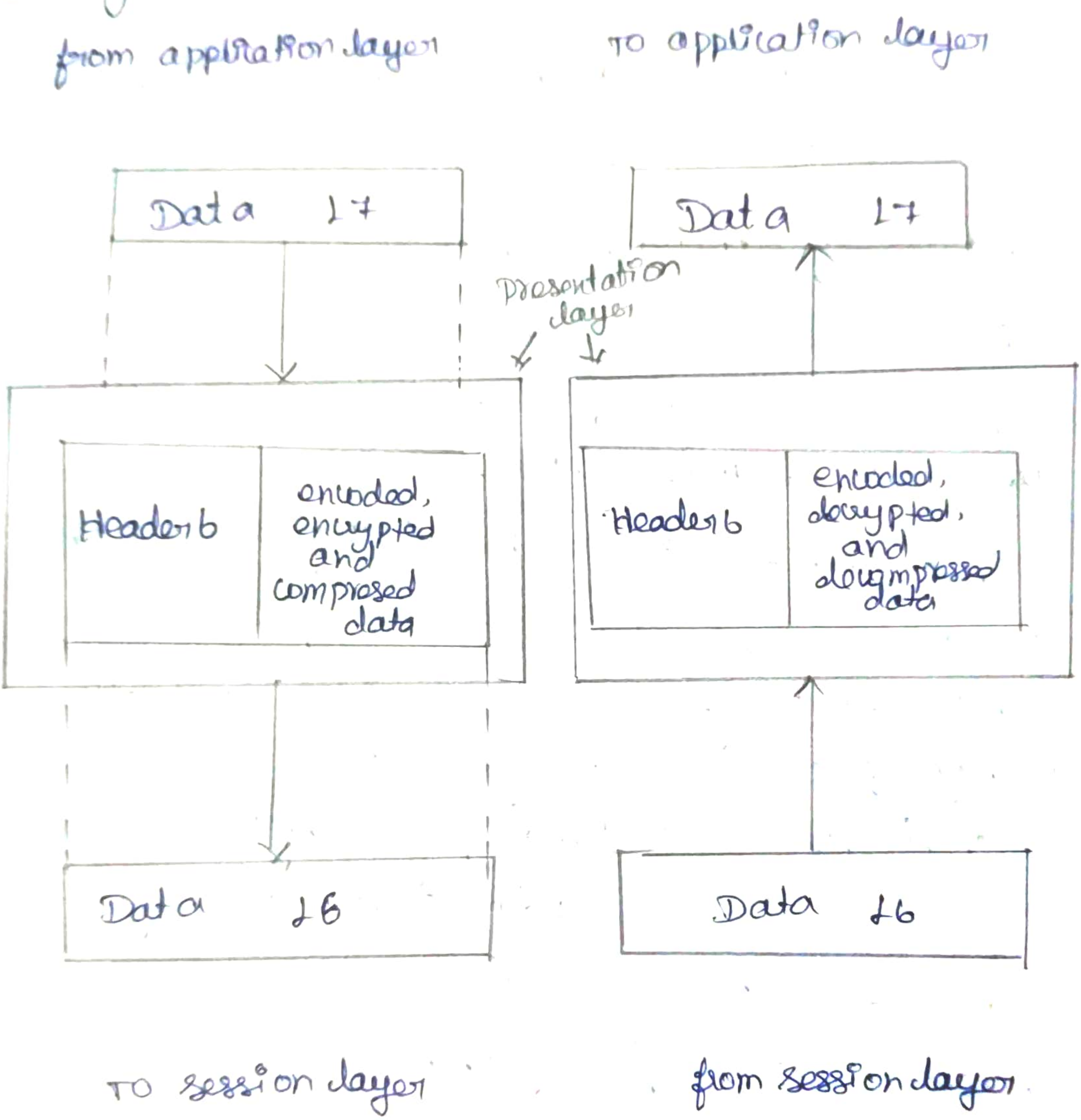
5. Session layer:

The session layer is network dialog controller ie it established and synchronizes the interaction between communication system.



6. Presentation layer :

The presentation layer deals with syntax and semantics of the information being exchanged.



7. Application layer:

Application layer is responsible for accessing the network by user. It provides user interfaces and other supporting services such as e-mail, remote file access, file transfer, sharing database, message handling (X.400), directory services (X.500)

Functions of application layer:

i) Network virtual terminal: It is a software version of that allows a user to lonto a remote host.

ii) File transfer, access and Management (FTAM): FTAM allows user to access files in remote host, to retrieve files and to manage files in remote computer.

iii) Mail services: E-mail forwarding, storage are the services under this category.

iv) Directory services: Directory services include access for global information and distributed database.

2) Explain the different ways to address the framing problem.

Framing problem:

The framing problem refers to the problem of identifying the beginning and end of each frame when a continuous stream of bits is transmitted between two devices.

The data link layer receives a stream of bits from the physical layer. It must divide this stream into meaningful units called frames.

For example:

1010101101010010111101010....

The receiver needs to determine:

[Frame 1] [Frame 2] [Frame 3]

Need for framing:

*Divides a bit stream into manageable units.

* Helps the receiver identify the start and end of data.

* provides synchronization between sender and receiver.

* Allows error detection.

* supports flow control and reliable delivery.

* Helps identify source and destination addresses

ways to address the framing problem:

The major framing techniques are:

1. character count
2. character stuffing
3. Bit stuffing
4. Flag bytes with physical layer coding violations.

Character count :

In the character count method, a field in the frame header specifies the number of characters / bytes in the frame.

Example :

Suppose the frame contains 5 characters.

The sender transmits.

5 | A B C D E

The receiver reads the first value 5 and and knows that the next five characters belong to the frame.

Working :

1. sender calculates the length of frame.
2. The length is placed in the header
3. Receiver reads the length field.
4. Receiver collects the specified number of bytes.
5. The frame ends after the specified number of bytes.

Character stuffing :

Character stuffing is also called byte stuffing

It is used when special characters are used to mark the beginning and end of a frame.

A special character called a flag or delimiter is placed at the start and end of the frame.

Example :

FLAG - framing boundary

ESC - escape character

A normal frame can be :

FLAG ABCDE FLAG.

Bit stuffing :

Bit stuffing is used in bit-oriented Protocols.

A special bit pattern called a flag is used to mark the beginning and end of a frame.

A commonly used flag pattern is

01111110

Flag bytes with physical layer

Another approach is to use special signal patterns that cannot normally occur in valid data.

In some physical layer encoding schemes, certain signal patterns are not used for normal data transmission.

These unused patterns can be used to indicate:

* Start of frame
* End of frame
* frame boundaries

Importance of framing:

Framing provides several important functions at the data link layer.

1. synchronization - It allows the receiver to identify frame boundaries.

2. Error Detection - Frames provide a suitable unit for error detection using techniques such as CRC.

3. Addressing - Data link addresses can be included in the frame header

4. Flow control - Frames allow the receiver to control the amount of data being transmitted.

5. Reliable Transmission - Individual frames can be acknowledged and retransmitted if necessary.

6. Efficient data transfer - large streams of data are divided into smaller manageable unit.

Applications:

* Robotics
* Autonomous vehicles
* Expert system
* AI planning
* Game AI
* Smart homes
* Multi-agent systems.

3) Explain about IPV6? compare IPV4 and IPV6.

IPV6:

* IPV4 provides the host to host communication between systems in the internet. IPV4 has played a central role in the internetworking environment for many years. It has proved flexible enough to work on the successor of IPV4 the many different networking technologies.

* In the early 1990 the IETF began to work on the successor of IPV4 that would solve the address exhaustion problem and other scalability problems.

Advantages of IPV6:

1. Larger address space
2. Better header format
3. security capabilities
4. New options
5. support for resource allocation
6. Allowance for extension

* IPV6 addresses are 128 bits in length. Addresses are assigned to individual interface on nodes, not to the node themselves. A single interface may have multiple unique unicast addresses. The first field of any IPV6 address is the variable length format prefix, which identifies various categories of addresses.

IPV6 addresses :

A new notation has been devised for writing 16-byte addresses. They are written as eight groups of four hexadecimal digits with colons between the groups, like this

8000 : 0000 : 0000 : 0000 : 0123 : 4567 : 89AB : CDEF

Optimization:

1. Leading zeros within a group can be omitted so 0123 can be written as 123

2 one or more groups of 16 zero bits can be replaced by a pair of colons. The address now becomes.

8000 : : 123 : 4567 : 89AB : CDEF

IPV4 and IPv6 :

IPV4 :

1. Header size is 32 bits
2. It cannot support autoconfiguration.
3. Cannot support realtime application.
4. No security at network layer
5. Throughput and delay is more.

IPv6 :

1. Header size is 128 bits
2. supports auto configuration
3. supports realtime application
4. provides security at network layer
5. Throughput and delay is less.